

Functional Analysis — Homework 1

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Problem 1. For $p \in [1, \infty)$, give an example of a sequence which is in ℓ^q for all $q > p$, but not in ℓ^p .

Proof. Consider the sequence (x_n) defined by

$$x_n = \left(\frac{1}{n}\right)^{\frac{1}{p}}.$$

Then $\|(x_n)\|_p^p = \sum_1^\infty 1/n$, which is the harmonic series, so $(x_n) \notin \ell^p$, but for any $q > p$ we have $(1/n)^{q/p} < 1/n$ when $n \neq 1$. Thus $\|(x_n)\|_q^q$ converges and $(x_n) \in \ell^q$. $\square$

Problem 2. If a subspace S of a normed space V contains a nonempty open set, then $S = V$.

Proof. Let U be a nonempty open subset of S , let $p \in U$, and let $\delta > 0$ such that $B_\delta(p) \subset U$. Now consider an arbitrary $x \in V$ and $\alpha > 0$. The idea is to “nudge p in the direction of x ” while remaining in $B_\delta(p)$ by finding α such that $\alpha(x - p) + p$ is close enough to p . Indeed,

$$\|[\alpha(x - p) + p] - p\| = \|\alpha(x - p)\| = \alpha\|x - p\|,$$

so by choosing $\alpha < \delta/\|x - p\|$, we get $\alpha(x - p) + p \in B_\delta(p) \subset S$. We can now use the closure of S under addition and scalar multiplication to get that

$$[\alpha(x - p) + p] + [\alpha p - p] = \alpha x \in S,$$

and thus multiplying by $1/\alpha$ we get $x \in S$. Since x was arbitrary, we have shown that $V \subset S$, and thus $V = S$. $\square$

Problem 3. Show that in any normed space V , the operations

$$+ : V \times V \rightarrow V \quad \cdot : \mathbb{R} \times V \rightarrow V$$

of addition and scalar multiplication are continuous.

Proof. First we do scalar multiplication. Let $(x_n) \rightarrow x$ in V and $(c_n) \rightarrow c$ in $\mathbb{R}$. Then if

Now let $(x_n) \rightarrow x$ and $(y_n) \rightarrow y$ in V . Then given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ (take such an integer for (x_n) and one for (y_n) and then N is the max of these two) such that when $n > N$,

$$\begin{aligned} \frac{\varepsilon}{2} &> \|x - x_n\| \geq \|x\| - \|x_n\| \\ \frac{\varepsilon}{2} &> \|y - y_n\| \geq \|y\| - \|y_n\|. \end{aligned}$$

UNFINISHED $\square$

Problem 4. Let V be a normed space. Show that a linear map $f : V \rightarrow \mathbb{R}$ is continuous if and only if $\ker(f)$ is closed.

Proof. If f is continuous, then since $\ker(f)$ is the preimage of the closed set $\{0\}$, $\ker(f)$ is closed. If $\ker(f)$ is closed, let (x_n) be a sequence converging to x in V .

... Thus $(f(x_n)) \rightarrow f(x)$, so f is continuous as desired. **UNFINISHED**

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Problem 5. Let $\rho : V \rightarrow [0, +\infty)$ be a seminorm. Prove that if there exists a positive number $\delta > 0$ such that the preimage $\rho^{-1}[0, \delta]$ contains an open ball, then ρ is continuous.

Proof. **UNFINISHED**

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Problem 6. Let V be a normed space. Show that V is Banach if and only if every absolutely convergent series is convergent.

Proof. First, assume V is Banach and $\sum \|x_i\| < \infty$. Then $(\|x_i\|)$ is Cauchy in $\mathbb{R}$, so given $\varepsilon > 0$, we can choose $N > 0$ such that when $m > n > N$,

$$\sum_1^m \|x_i\| - \sum_1^n \|x_i\| = \sum_{n+1}^m \|x_i\| < \varepsilon.$$

Then we also have that

$$\left\| \sum_1^m x_i - \sum_1^n x_i \right\| = \left\| \sum_{n+1}^m x_i \right\| \leq \sum_{n+1}^m \|x_i\| < \varepsilon,$$

so the partial sums of $\sum_1^\infty x_i$ are also Cauchy and thus converge since V is Banach.

Now if every absolutely convergent series is convergent, then let (x_i) be a Cauchy sequence in V . Define $y_1 := x_1$, and then recursively define $y_n := x_n - \sum_{i=1}^{n-1} y_i$ for all $n > 1$, so that $x_n = \sum_1^n y_i$ for all n . We then just need to show that $\sum y_i$ is absolutely convergent, and we will have shown that x_n converges and thus V is Banach.

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Problem 7. In this problem, you will show that for any number $p > 1$, the function

$$\|x\|_p = \|(x_1, \dots, x_n)\|_p = (|x_1|^p + \dots + |x_n|^p)^{1/p}$$

is a norm on $\mathbb{R}^n$. The extension to ℓ^p follows easily from that. It will be helpful to define a number q , called the conjugate exponent to p , by the equation: $\frac{1}{p} + \frac{1}{q} = 1$.

- (a) (Young's inequality) Use the fact that $\ln(x)$ is a concave function to prove that if a, b are positive numbers, then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

- (b) (Hölder's inequality) Given any $x, y \in \mathbb{R}^n$, use Young's inequality to prove that

$$|\langle x, y \rangle| \leq \|x\|_p \|y\|_q$$

where $\langle x, y \rangle$ denotes the standard dot product in $\mathbb{R}^n$.

(c) (Minkowski's inequality) Given any $x, y \in \mathbb{R}^n$, use Hölder's inequality to prove

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p,$$

which is the triangle inequality for $\|\cdot\|_p$.

Proof.

(a) Since $\ln(x)$ is concave and the tangent line at x_0 is

$$\begin{aligned} y &= \frac{1}{x_0}(x - x_0) + \ln(x_0) \\ &= \frac{x}{x_0} - 1 + \ln(x_0) \dots \end{aligned}$$

(b)

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